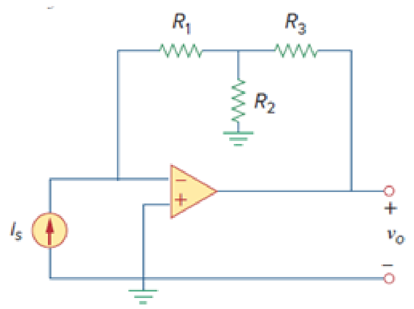
Problem

(a) Determine the ratio *vo/is* in the op amp circuit of Fig. 5.54.

(b) Evaluate the ratio for *R*1 = 20 kΩ, *R*2 = 25 kΩ, *R*3 = 40 kΩ.

Figure. 5.54:



Step-by-step solution

Step 1 of 3

(a)

Let V_1 be the voltage at the node where the three resistors meet.

Applying KVL, at this node gives,

$$i_S = \frac{V_1}{R_2} + \frac{V_1 - V_0}{R_3}$$

$$i_S = V_1\left[\frac{1}{R_2} + \frac{1}{R_3}\right] - \frac{V_0}{R_3} \text{.........................} (1)$$

At the inverting terminal,

$$i_S = \frac{0 - V_1}{R_1}$$

$$V_1 = -i_S R_1 \text{..........................} (2)$$

Comments (1)

Anonymous

Applying K[C]L at this node...

Step 2 of 3

Combining (1) and (2) leads to,

$$i_S\left[1 + \frac{R_1}{R_2} + \frac{R_1}{R_3}\right] = \frac{-V_0}{R_3}$$

$$\boxed{\frac{V_0}{i_S} = -\left[R_1 + R_3 + \frac{R_1 R_3}{R_2}\right]}$$

Step 3 of 3

(b)

Given that,

$R_1 = 20\text{k}\Omega$,

$R_2 = 25\text{k}\Omega$ and

$R_3 = 40\text{k}\Omega$.

Therefore,

$$\frac{V_0}{i_S} = -\left[R_1 + R_3 + \frac{R_1 R_3}{R_2}\right]$$

$$\frac{V_0}{i_S} = -\left[20\times10^3 + 40\times10^3 + \frac{20\times10^3\times40\times10^3}{25\times10^3}\right]$$

$$\boxed{\frac{V_0}{i_S} = -92\text{k}\Omega}$$